Yanni Mei

Ph.D Student in Human-Computer Interaction

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Research Interests

My research focuses on the intersection of Human-Computer Interaction and Extended Reality, trying to understand what happens when AR becomes part of everyday life: how it fits into the small, ordinary moments, how it changes the way we interact with technology, and how it reshapes our relationship with the world around us.

Education

PhD, Human-Computer Interaction 2023.2 - present
TU Darmstadt, Germany
Advisor: Prof. Jan Gugenheimer

MSc, Design For Interaction 2018 – 2020
TU Delft, the Netherlands

BEng in Environmental Design 2013 – 2017
Tongji University, China

Publications

[5] **Mei, Y.**, Wendt, S., Jombacher, J., Müller, F., & Gugenheimer, J. (2026,). ShadAR: Real-time Prompting On-demand Visual Augmentation in Augmented Reality. Conditionally Accepted by UIST’26.

[4] **Mei, Y.**, Wendt, S., Müller, F., & Gugenheimer, J. (2026). Meme, Myself and AR: Exploring Memes Sharing in Face-to-face Conversation using Augmented Reality. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems.

[3] **Mei, Y.**, Wombacher, J., Tseng, W. J., Krauß, V., & Gugenheimer, J. (2026). Embodied Spatial Simulation: Enabling Embodied Interaction with Human Motion Simulation in Extended Reality. IEEE Transactions on Visualization and Computer Graphics.

[2] Li, J., Subramanyam, S., Jansen, J., **Mei, Y.**, Reimat, I., Lawicka, K., & Cesar, P. (2021, October). Evaluating the user Experience of a Photorealistic Social VR Movie. In ISMAR (pp. 284-293).

[1] **Mei, Y.**, Li, J., De Ridder, H., & Cesar, P. (2021). Cakevr: A social virtual reality (vr) tool for co-designing cakes. In Proceedings of the 2021 CHI conference on human factors in computing systems.

Demonstrations

ShadAR: LLM-driven shader generation to transform visual perception in Augmented Reality
Mei, Y., Wendt, S., Jombacher, J., Müller, F., & Gugenheimer, J. *CHI’26*

ShadAR: LLM-driven shader generation to transform visual perception in Augmented Reality
Mei, Y., Wendt, S., Müller, F., & Gugenheimer, J. *ISMAR’25*

Mediascape XR: A Cultural Heritage Experience in Social VR
Reimat, I., **Mei, Y.**, Alexiou, E., ... & Cesar, P. *MM’22*

Research Experience

Junior Researcher (FTE) 2021

Centrum Wiskunde & Informatica (CWI) / Distributed & Interactive Systems group

Designed and evaluated a SocialVR system for remote cultural heritage experiences. The project resulted in a demo paper (MediaScape) presented at MM'22, which received the Best Demo Award.

Master Thesis Intern 2020

Centrum Wiskunde & Informatica (CWI) / Distributed & Interactive Systems group

Designed, developed, and evaluated a SocialVR system for remote co-design practices. The project led to a full paper (CakeVR) at CHI'21.

Industry Experience

UX Researcher 2022

BYD Auto Europe, the Netherlands

Designed and developed a VR driving simulator to facilitate user research across teams and with clients, to inform vehicle redesign requirements.

UX Design Intern 2018

Accenture, China

Generated UX design concepts and crafted user interfaces for smart home appliances. Facilitated creative workshops with clients.

Teaching

Teaching Assistant — Human Computer Interaction Spring Semester 2025

TU Darmstadt

VR Prototyping Exercise Creation, Written Exam Creation, Grading

Thesis Supervision

Bachelor Thesis — Oliver Hönig 2026

TU Darmstadt

Proactive AI Moderation for Goal-Oriented Online Meetings

Master Thesis — Lars Heublein 2024

TU Darmstadt

Augmented Reality for Urban Planning: Integrating Data Visualization with Physical Models

Awards

- **2022** — Best Demo Award in ACM Multimedia
- **2016** — Chinese Government Scholarship for Undergraduates (2 out of 110)
- **2015** — Tongji University First Class Scholarship for Undergraduates (Top 1%)
- **2014** — Tongji University Second Class Scholarship for Undergraduates (Top 5%)

Skills

Programming/Tool: Python, C#, HTML&CSS, LaTex, Blender3D, Figma
Languages: English (fluent), Mandarin (Native), German (Daily Conversation)